Metadata-Driven Intelligent News Article Retrieval System

With Retrieval-Augmented Answer Generation (RAG)

ZEROTH REVIEW

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The Information Retrieval Problem

Digital news growth creates vast daily article volumes across domains. Traditional systems face key challenges:

- Failure to capture deep contextual relevance.
- Rich metadata (source, date, tags) is ignored.
- Simple ranked lists lack meaningful synthesis.

Core Project Objectives

- Design metadata-aware retrieval systems.
- Implement IR models (TF-IDF/BM25).
- Analyze Top vs Bottom ranked documents.
- Apply RAG for grounded answer synthesis.
- Use small-scale LLMs for organization.
- Develop a standalone, explainable AI system.

News Articles Corpus Overview

Content

Full article text, summaries, and domain-specific titles.

Metadata

Categories, publication dates, sources, and keywords.

Sources

Open-source news corpora and public news datasets.

Proposed Methodology

A hybrid approach combining classical IR and RAG

Core Retrieval Pipeline

Preprocessing

Tokenization, normalization, and metadata filtering.

IR Ranking

TF-IDF or BM25 computation for document relevance.

Retrieval Set

Selection of Top 20 vs Bottom 20 for evaluation.

Classic IR Scoring (TF-IDF)

$$W_{i,j} = tf_{i,j} \times \log \frac{N}{df_i}$$

RAG & Small LLM Role

Synthesis & Organization

Retrieved documents act as the **exclusive context** for the Small LLM. This ensures answers are grounded and factual.

The LLM is used only **after retrieval** to remove redundancy and structure the presentation of ranked results.

Key Design Philosophy

— Project Core Implementation Strategy

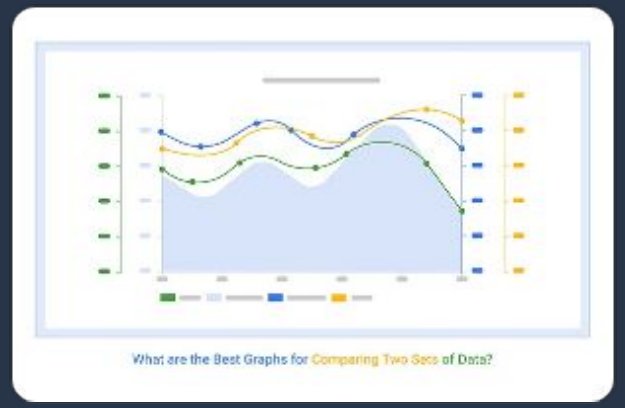
"Retrieval is performed using classical IR techniques, and the language model is used only for post-retrieval organization and summarization."

— Project Core Implementation Strategy

What Makes This Distinct

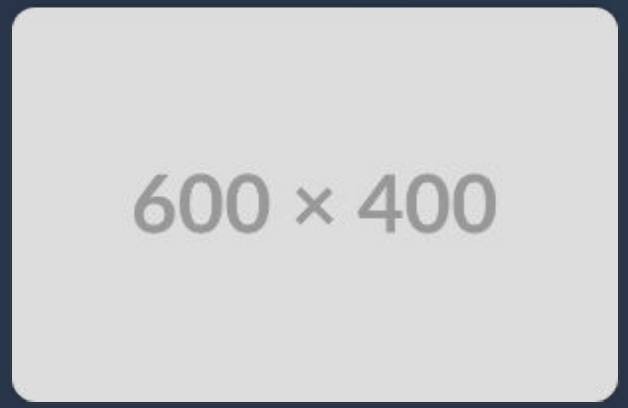
Metadata Aware

Ranking improved via date, source, and tag weighting.



Comparison

Evaluation of Top vs Bottom ranked document content.



Standalone

Reproducible system using small, local language models.

Expected Project Outcomes

100%

Grounded Answers

Reliable IR Deliverables

- High-accuracy retrieval of relevant news articles.
- Concise, metadata-aware summaries via RAG.
- Transparent pipeline demonstration for XAI.
- Standalone reproducible code-base for IR systems.